

Malware Dynamic Analysis Fundamentals using AnyRun.

This Lab aided in simulating the workflow of a junior SOC Analyst or Malware Analyst by analyzing malicious files, investigating alerts, and documenting findings using tools and techniques learned from the LetsDefend “Malware Analysis Fundamentals” course.

AnyRun is an interactive malware analysis sandbox designed for rapid threat investigation. The platform offers both free and paid tiers. While the free version allows basic analysis

Requirements:

- Windows/Linux VM (for lab replication)
- Tools:
- **VirusTotal**
- **Any.Run / Hybrid Analysis**
- **Strings**
- **PEStudio**
- **Procmon, Wireshark** (optional)

Let's dive in

Malware Analysis Observations

During manual execution, the malware spawned two child processes:

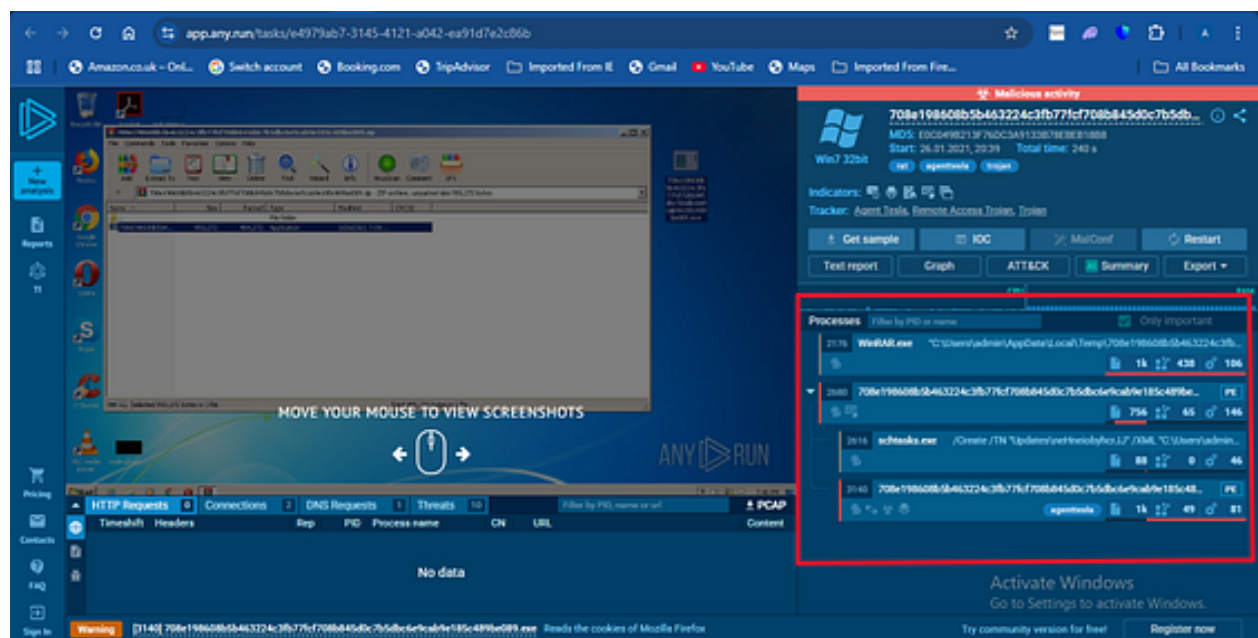
1. **schtasks.exe** — Establishes persistence by creating a scheduled task.
2. “*AgentTesla*” — Identified as malicious by AnyRun’s analysis.

- Navigate to the *Processes* tab to view detailed information in the red-highlighted panel.
- We will analyze each process systematically, excluding WinRAR.exe (launched during malware extraction and unrelated to malicious activity).

Process ID 2680 Details

Clicking on this process displays its attributes in the red panel, including:

- Loaded modules
- Network activity
- Registry modifications



AnyRun Interface

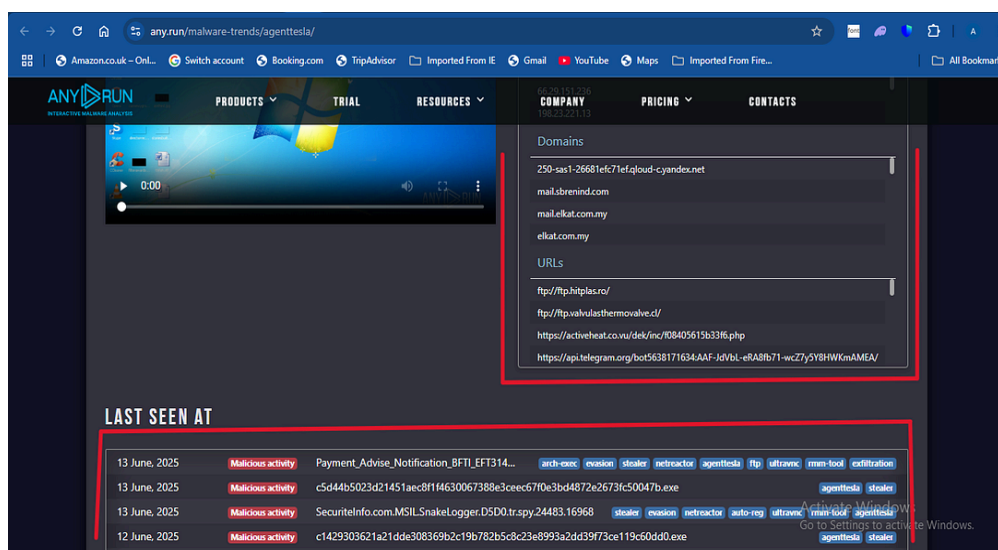
Process Analysis Details

The “*More Info*” button provides access to comprehensive process details. For Process **ID 2680**, our investigation reveals the malware:

1. Task Scheduler Utilization — Creates scheduled tasks to maintain persistence
2. Suspicious File Operations:
 - Deploys an executable with an anomalously old compilation timestamp
 - Generates multiple files in the user directory, suggesting payload deployment or data collection

This behavior pattern is consistent with common malware persistence and operational tactics.

While you explore the other options in the process panel for further information, you’d notice the summary tab. When you click on it, you’re redirected to this section below, where you have extended details of **Agent Tesla**



Agent Tesla Detailed Report

Process Analysis (ID: 2616 — *schtasks.exe*)

- Process Identification: Confirmed as Task Scheduler component (*schtasks.exe*)
- Command Line Analysis:
- Creates scheduled task named “Updates\neHneiobyhcrJJ”
- Utilizes configuration file “**tmp5383.tmp**”
- Task Configuration Examination:
- Scheduled task executes “*neHneiobyhcrJJ.exe*”
- File location and additional parameters visible in configuration

app.any.run/tasks/e4979ab7-3145-4121-a042-ea91d7e2c86b

Amazon.co.uk - Onl... Switch account Booking.com TripAdvisor Imported From IE Gmail YouTube Maps Imported From Fire...

Malicious activity

708e198608b5b463224c3fb77fc7f08b845d0c7b5dbdc6cabe...
MD5: EDC0498213F76DC3A9133878EBB81889
Start: 26.01.2021, 20:39 Total time: 240 s

Indicators: Get sample IOC MalConf Restart
Read more about Agent Tesla

Processes: Filter by PID or name

Process details ID 2680 Malicious

708e198608b5b463224c3fb77fc7f08b845d0c7b5dbdc6cabe...
Agent
Username: admin
Start: 416172ms Indicators:
Command line
"C:\Users\admin\Desktop\708e198608b5b463224c3fb77fc7f08b845d0c7b5dbdc6cabe185c489a089.exe"

Danger 3
T1053.005 Scheduled Task (1)
Users Task Scheduler to run other applications

Activate Windows
Go to Settings to activate Windows.

Try community version for free! Register now

app.any.run/tasks/e4979ab7-3145-4121-a042-ea91d7e2c86b

Amazon.co.uk - Onl... Switch account Booking.com TripAdvisor Imported From IE Gmail YouTube Maps Imported From Fire...

Malicious activity

708e198608b5b463224c3fb77fc7f08b845d0c7b5dbdc6cabe9...
MD5: EDC0498213F76DC3A9133878EBB81889
Start: 26.01.2021, 20:39 Total time: 240 s

Indicators: Get sample IOC MalConf Restart
Text report Graph Read more about Agent Tesla Export

Processes: Filter by PID or name

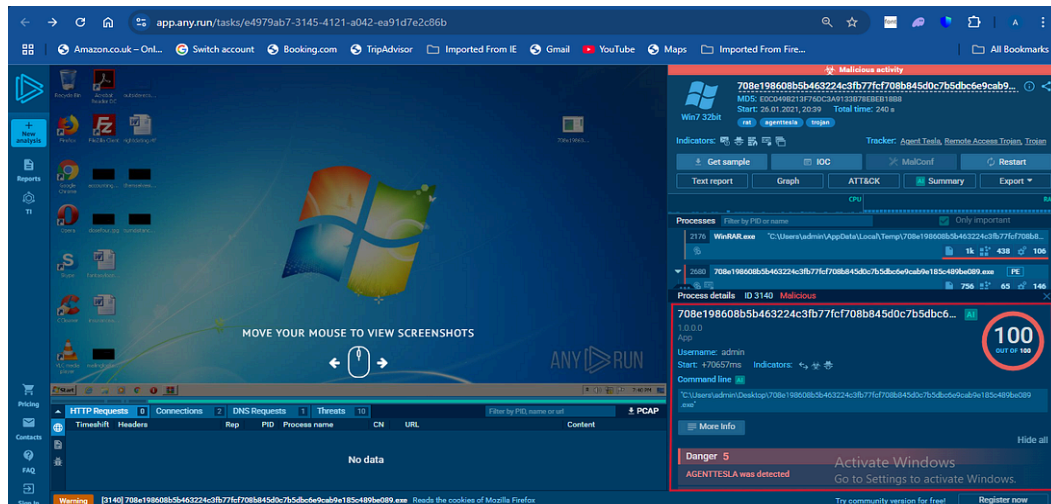
Process details ID 2616 No verdict

schtasks.exe
6.1.7600.16385 (win7_rtm.090713-1255)
Manages scheduled tasks
Username: admin
Start: 470516ms
Command line
"C:\Windows\System32\schtasks.exe" /Create /TN "Updates\nehtweidyhoch.LJ" /XML "C:\Users\admin\AppData\Local\Temp\imp5383.tmp"

Danger 1
T1053.005 Scheduled Task (1)
Loads the Task Scheduler COM API

Activate Windows
Go to Settings to activate Windows.

Try community version for free! Register now



Assessing the information under the processes section

When we examine the process with ID **3140**:

- This malware is recognized by AnyRun as **AgentTesla**.
- Steals credentials,
- Creating files in the user directory

In order to further your investigation as a junior SOC analyst as regards this lab, navigate to the lower left panel, under the Windows screen, where you'd find different tabs including HTTP Request, Connections, etc. Once you check the connections tab, more information is displayed, where you can see ports, traffic, etc. To save time, you find two traffics in the panel, the one with the higher amount of data traffic (988 b). Click on it and you are able to view the network stream.

The screenshot displays the ANY.RUN web-based malware analysis platform. The main window shows a virtualized Windows 7 desktop with various icons and a taskbar. On the right side, a sidebar provides detailed information about the running processes. A specific process with ID 3140 is highlighted, showing its name as 'WebRAR.exe' and its path as 'C:\Users\admin\AppData\Local\Temp\708e198608b5b463224c3fb77cf708b845dc7b5dbdc6e9cab9'. The process details section indicates that 'AGENT TESLA was detected' and lists several actions: 'T1555.003 Credentials from Web Browsers (1)', 'T1552.001 Credentials in Files (2)', 'T1518 Software Discovery (1)', and 'Application was dropped or rewritten from another process'. A red box highlights the 'Process details' section, and another red box highlights the 'Connections' tab in the bottom left pane.

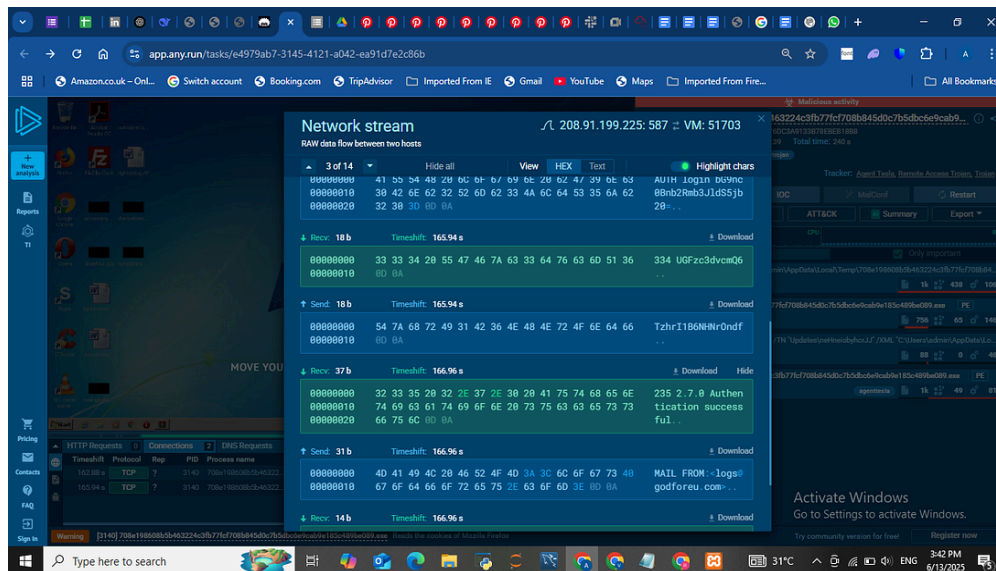
This screenshot shows the same ANY.RUN interface, but with the 'Connections' tab selected in the bottom left pane. This tab displays a table of network connections. The table has columns for 'TimeShift', 'Protocol', 'Process name', 'CN', 'IP', 'Port', 'Domain', 'ASN', and 'Traffic'. Two connections are listed, both originating from process ID 3140 and connecting to 'smtp.godfor...'. The first connection is a TCP connection to port 587, and the second is a TCP connection to port 587. The traffic volume for both connections is 988 b and 7 Kb respectively. A red box highlights the 'Connections' tab and the table of connections. Another red box highlights the 'Process details' section in the right sidebar, which shows the same information as the first screenshot.

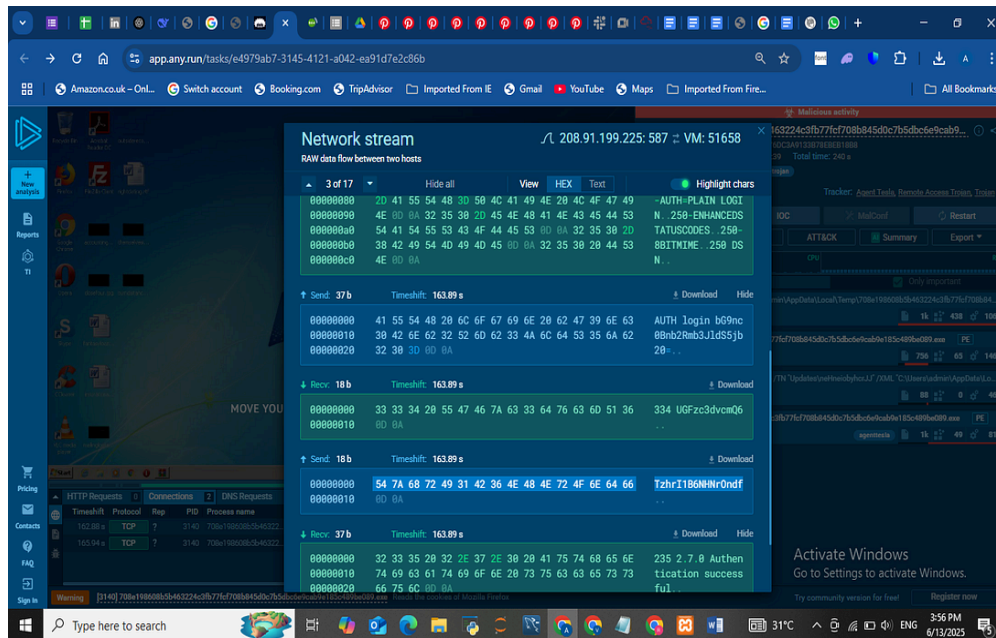
Under the process with ID 3140 is found more info

Here, in the network stream, you see view (hex & text). For clarity, you can choose text, then follow the stream and read through the content. The green

highlighted part is the *received data*, while the blue highlighted part is the *sent data*. Hence, there's an established communication. After reading through the communication process, a few things are observed.

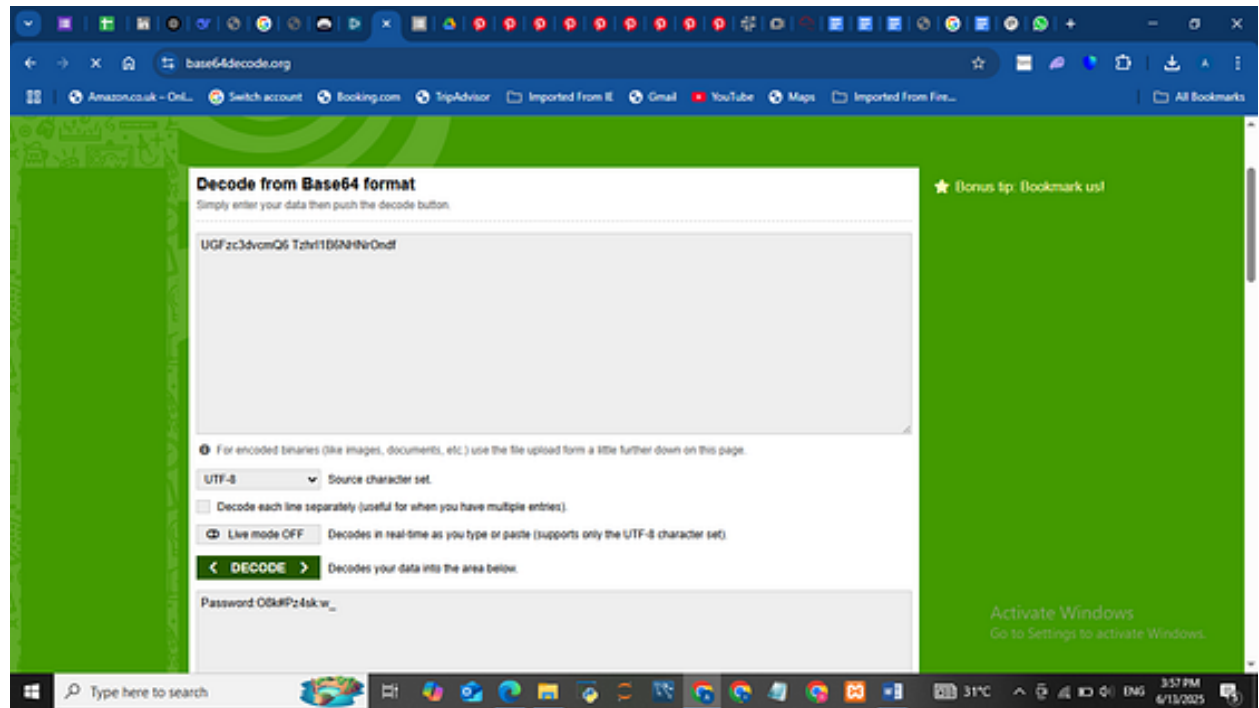
- The content encoding mentioned is *Base64*
- There was a request for *Auth Login*, meaning the user is to input their authentication information before gaining access.
- A corresponding Authentication information then followed in an encrypted format, but we already know the encoding that's in use.





Investigate the PCAP file

- Hence, we decode the encrypted content using an online Base64 decoder. Afterwards, we get the plain text that the user typed in to gain access to the system.



Base64 encoding/decoding website

And that sums up this part of the lab. You have found the flags and now understand the basic niceties of analyzing malware dynamically.